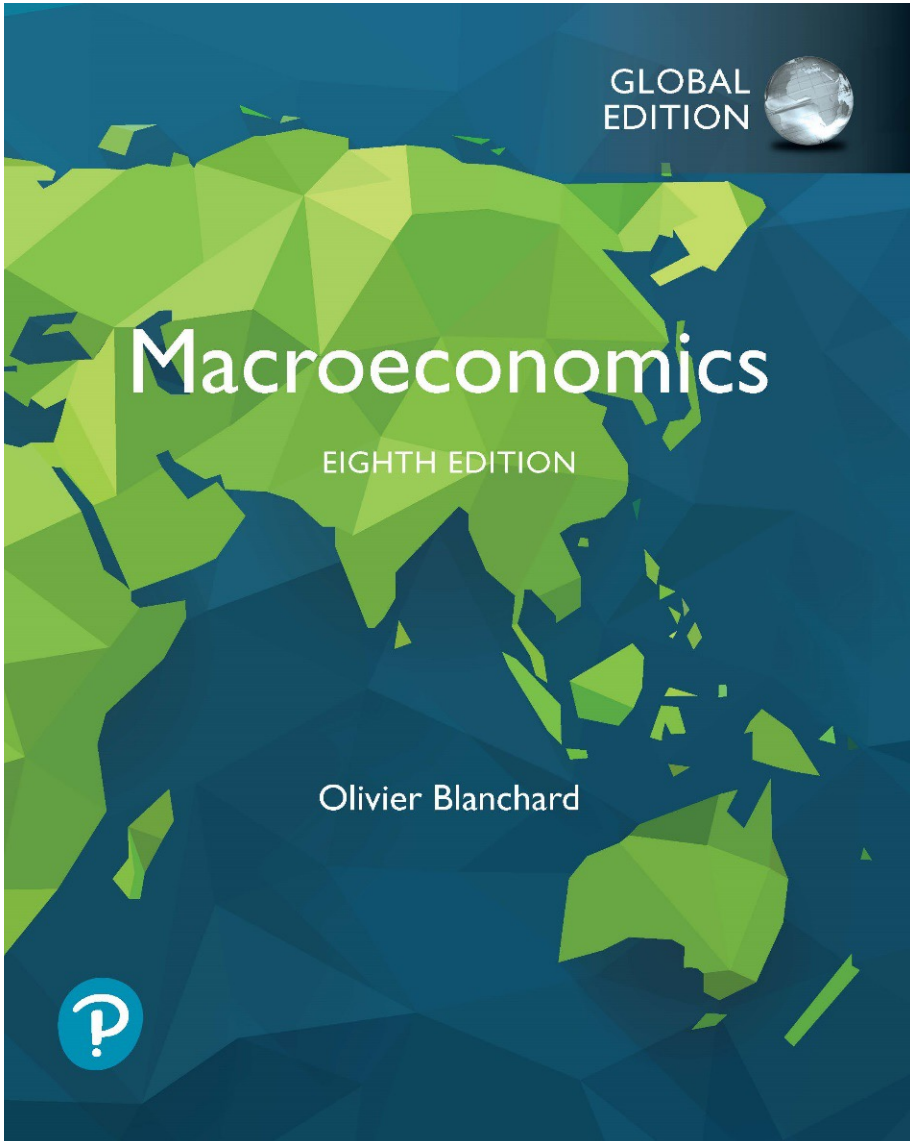
Macroeconomics

Eighth Edition, Global Edition



Chapter 11

Saving, Capital Accumulation, and Output

Pearson

Chapter 11 Outline

Saving, Capital Accumulation, and Output

Saving, Capital Accumulation, and Output

- Since 1970, the U.S. saving ratio—the ratio of saving to gross domestic product—has averaged only 17%, compared to 23% in Germany and 29% in Japan.
- Even if a lower saving rate does not permanently affect the growth rate, it does affect the level of output and the standard of living.

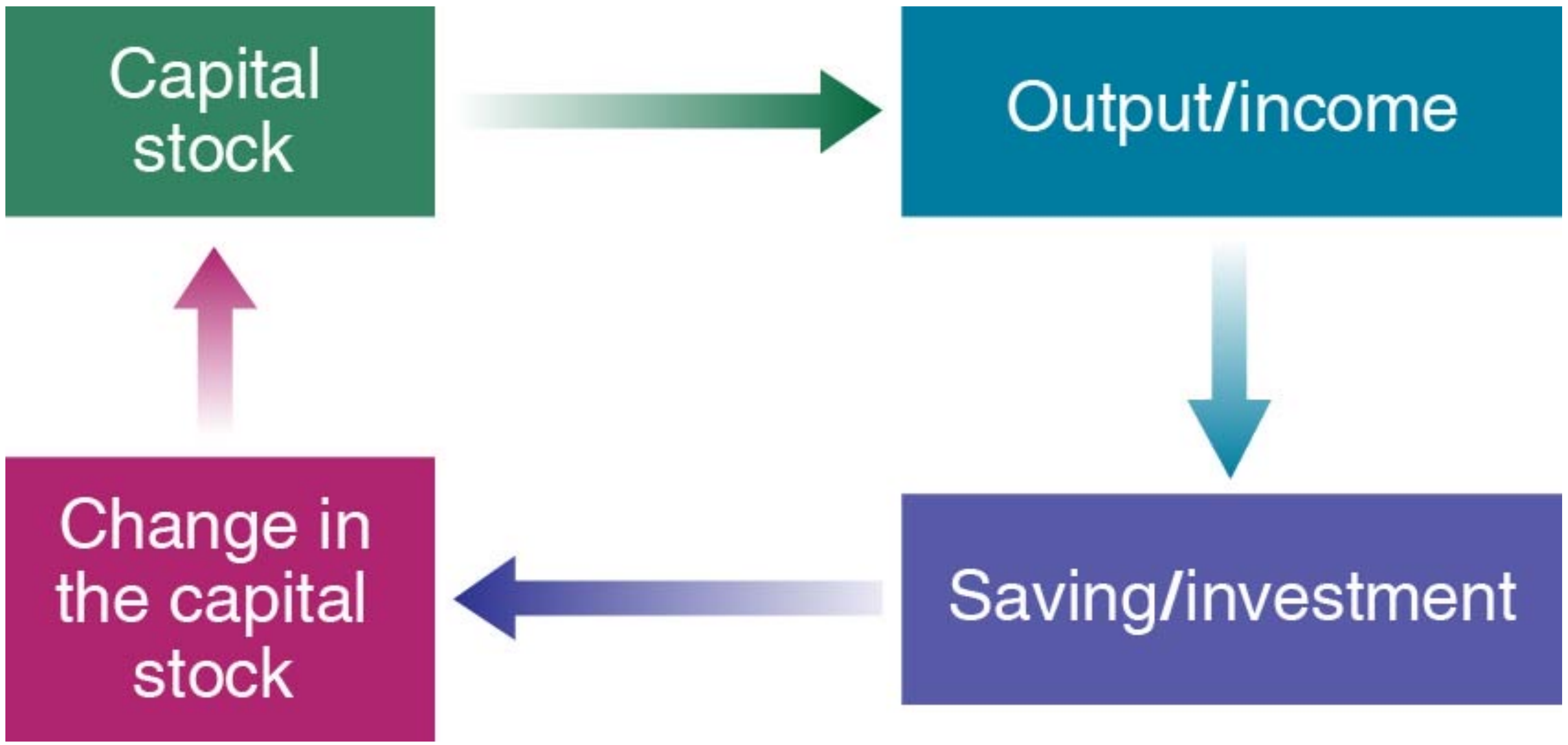
11.1 Interactions between Output and Capital (1 of 5)

- Output in the long run depends on two relations:
  - The amount of capital determines the amount of output being produced
  - The amount of output produced determines the amount of saving, which in turn determines the amount of capital being accumulated over time

Pearson

11.1 Interactions between Output and Capital (2 of 5)

Figure 11.1 Capital, Output, and Saving/Investment



The Harrod–Domar model

- The Harrod–Domar model is a Keynesian model of economic growth. It is used in development economics to explain an economy's growth rate in terms of the level of saving and of capital. It suggests that there is no natural reason for an economy to have balanced growth.

p.352

assume: $I = S$ 且 $S = \beta Y$

資本產出比為一常數 $\Rightarrow \frac{K}{Y} = \alpha$

沒有資本折舊

$\Delta K = I$

$\Rightarrow \Delta Y = \frac{1}{\alpha} \cdot \Delta K = \frac{1}{\alpha} \cdot I = \frac{1}{\alpha} \cdot S$

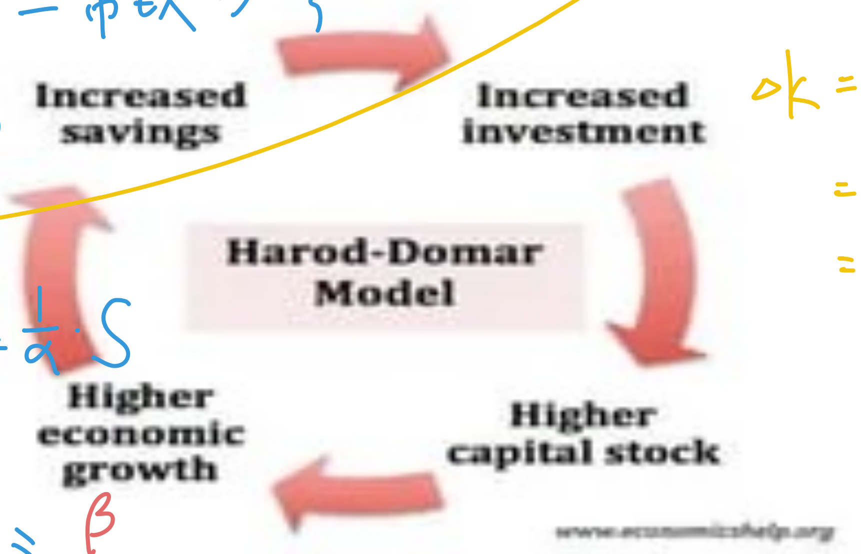
$\frac{\Delta Y}{Y} = \frac{1}{\alpha} \cdot \frac{I}{Y} = \frac{1}{\alpha} \cdot \frac{S}{Y} = \frac{\beta}{\alpha}$

Solow Model

$\Delta k = i - (\delta + n)k$

$= s - (\delta + n)k$

$= \beta f(k) - (\delta + n)k$



The Harrod–Domar model

- Assumptions

β：儲蓄率

1) I=S

① $Y = \min(\lambda_1 K, \lambda_2 L)$

2) $I = \Delta K$

3) K/Y=θ=constant. ($\Delta K/K = \Delta Y/Y$), (i.e., MP_K is constant)

s=S/Y, $\Rightarrow \beta Y$

I=sY= ΔK

$\Delta Y/Y$= $\Delta K/K$= sY/K=s/θ

Increase saving rate will increase growth rate of output

If θ=3, then it requires 30% of saving rate to having 10% economic growth rate. (e.g., Asian four Tigers)

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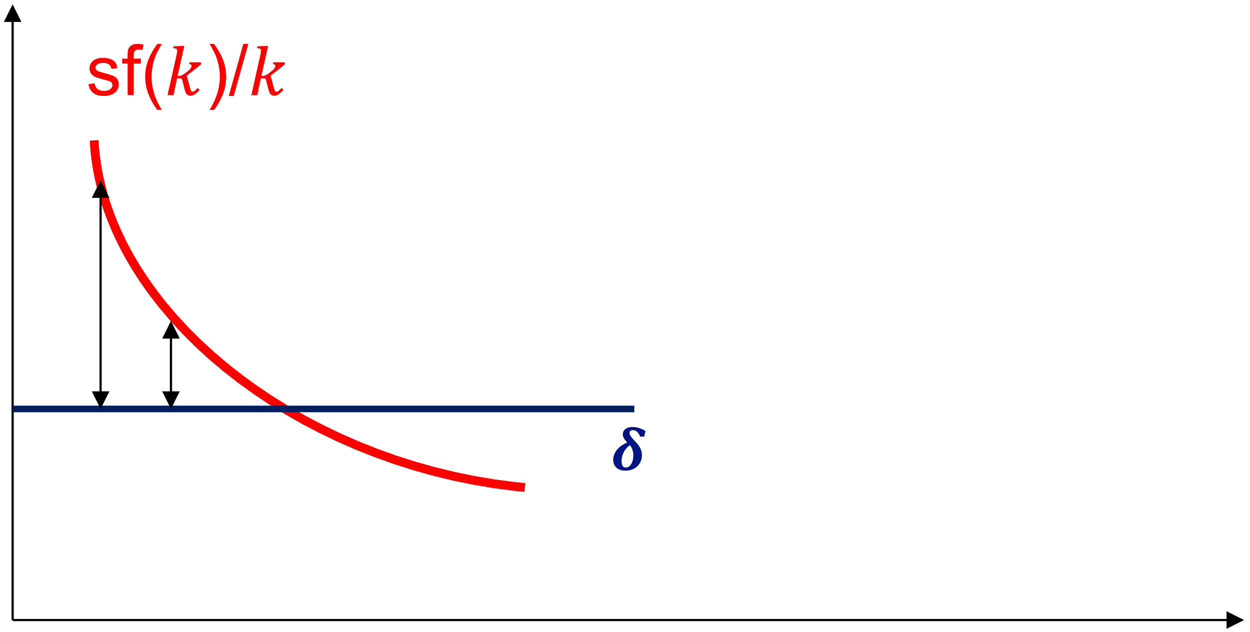
Neoclassical (Solow) Growth Model

- Max U(C) st Y=Y* →C=bY or S=sY and s=1-b
- Y=F(K,N) with C.R.S (MP_K is diminishing)
- $\Delta \mathrm{K} = \mathrm{I}\text{-}\delta K$ (δ is the depreciation rate of capital)
- I=S (macro-equilibrium condition)
- ΔK =sF(K,N)- δK
- Δk =sf(k)- $\delta k,\ where\ k = \frac{K}{Y}, y = \frac{Y}{N}$
- Implications: convergence hypothesis, poor countries grow faster than rich countries (catching-up); however, in the long run growth rate is zero!

Pearson

Catching-up process

- Δk =sf(k)- δk
- $\Delta k/k$ =sf(k)/k - δ (AP_k is diminishing)
- d($\Delta k/k$)/dk=s(f'(k)k-f(k))/k^2 < 0,
- i.e., growth rate of output is diminishing. ($\because$ $MP_k < AP_k$)



11.1 Interactions between Output and Capital (3 of 5)

- Recall Chapter 10:

$$\frac{Y}{N} = F\left(\frac{K}{N}, 1\right) \quad \text{or} \quad f\left(\frac{K}{N}\right) = F\left(\frac{K}{N}, 1\right)$$

- Assume that *N* is constant, and there is no technological progress, so *f* does not change over time:

$$\frac{Y_t}{N} = f\left(\frac{K_t}{N}\right) \qquad (11.1)$$

- Higher capital per worker leads to higher output per worker.

11.1 Interactions between Output and Capital (4 of 5)

- Assume:
  - The economy is closed: $I = S + (T - G)$
  - Public saving $(T - G)$ is 0: $I = S$
  - Private saving is proportional to income: $S = sY$
- So the relation between output and investment:

$$I_t = sY_t$$

- Investment is proportional to output.
- The higher (lower) output is, the higher (lower) is saving and so the higher (lower) is investment.

11.1 Interactions between Output and Capital (5 of 5)

- The evolution of the capital stock is:

$$K_{t+1} = (1-\delta)K_t + I_t$$

- Replace investment by the above expression and divide both sides by *N*:

$$\frac{K_{t+1}}{N} = (1-\delta)\frac{K_t}{N} + s\frac{Y_t}{N}$$

or

$$\frac{K_{t+1}}{N} - \frac{K_t}{N} = s\frac{Y_t}{N} - \delta\frac{K_t}{N} \qquad (11.2)$$

- The change in the capital stock per worker is equal to saving per worker minus depreciation.

11.2 The Implications of Alternative Saving Rates (1 of 10)

- Combining equations (11.1) and (11.2):

$$\frac{K_{t+1}}{N} - \frac{K_t}{N} = sf\left(\frac{K_t}{N}\right) - \delta\frac{K_t}{N} \qquad (11.3)$$

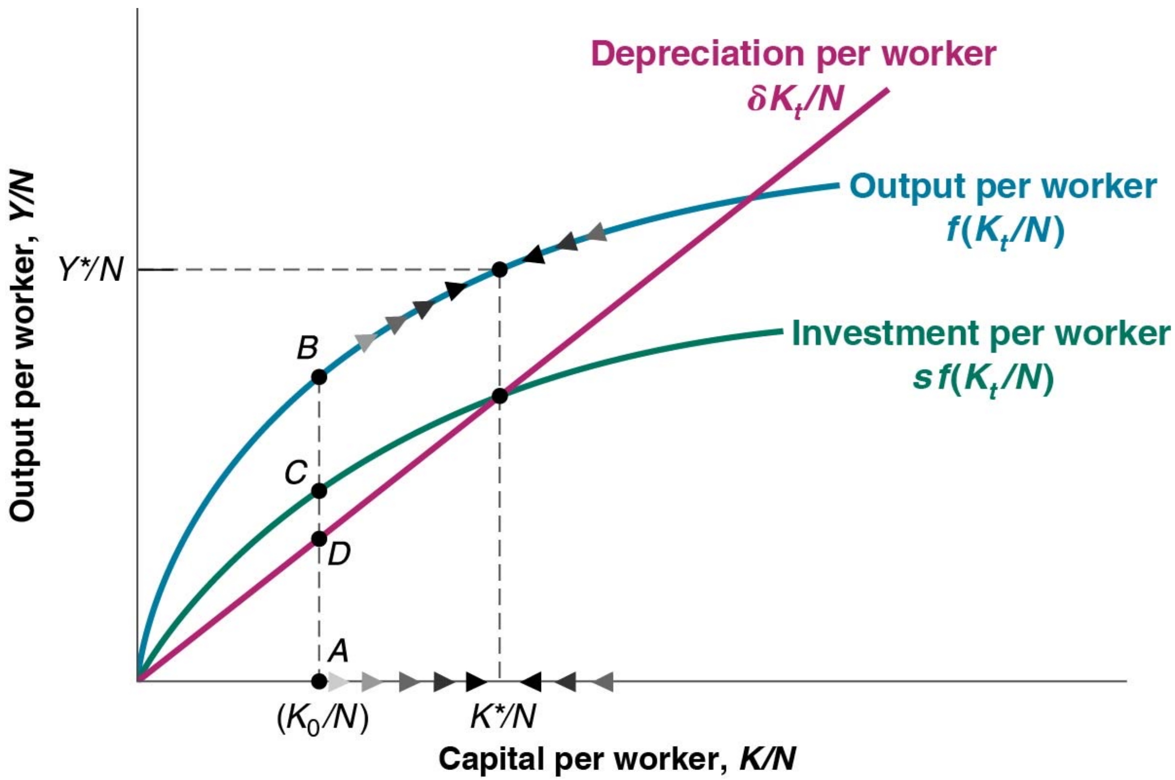
$$\underset{\text{from year } t \text{ to year } t+1}{\text{change in capital}} = \underset{\text{during year t}}{\text{Invesment}} - \underset{\text{during year t}}{\text{depreciation}}$$

- If investment per worker exceeds (is less than) depreciation per worker, the change in capital per worker is positive (negative).

11.2 The Implications of Alternative Saving Rates (2 of 10)

Figure 11.2 Capital and Output Dynamics

When capital and output are low, investment exceeds depreciation and capital increases. When capital and output are high, investment is less than depreciation and capital decreases.



11.2 The Implications of Alternative Saving Rates (3 of 10)

- The state in which output per worker and capital per worker are no longer changing is called the **steady state** of the economy.

$$s\, f\left(\frac{K\,*}{N}\right) = \delta \frac{K\,*}{N} \qquad (11.4)$$

- The steady-state value of capital per worker is such that the amount of saving per worker is sufficient to cover depreciation of the capital stock per worker.

$$\left(\frac{Y\,*}{N}\right) = f\left(\frac{K\,*}{N}\right) \qquad (11.5)$$

Focus: Capital Accumulation and Growth in France in the Aftermath of World War II

- France suffered heavy losses in capital when World War II ended in 1945.
- The growth model predicts that France would experience high capital accumulation and output growth for some time.
- From 1946 to 1950, French real GDP indeed grew at 9.6% per year.

Table 1 Proportion of the French Capital Stock Destroyed by the End of World War II

Railways	Tracks	6%	Rivers	Waterways	86%
	Stations	38%		Canal locks	11%
	Engines	21%		Barges	80%
	Hardware	60%	Buildings	(numbers)	
Roads	Cars	31%		Dwellings	1,229,000
	Trucks	40%		Industrial	246,000

Source: Gilles Saint-Paul, "Economic Reconstruction in France, 1945–1958," in Rudiger Dornbusch, Willem Nolling, and Richard Layard, eds. *Postwar Economic Reconstruction and Lessons for the East Today* (Cambridge, MA: MIT Press, 1993).

A Numerical Example

$\Delta k = i - \delta k$

Approaching the Steady State: A Numerical Example

Assumptions: $y = \sqrt{k}$; $s = 0.3$; $\delta = 0.1$; initial $k = 4.0$

Year	k	y	c	i	δk	Δk
1	4.000	2.000	1.400	0.600	0.400	0.200
2	4.200	2.049	1.435	0.615	0.420	0.195
3	4.395	2.096	1.467	0.629	0.440	0.189
4	4.584	2.141	1.499	0.642	0.458	0.184
5	4.768	2.184	1.529	0.655	0.477	0.178
⋮						
10	5.602	2.367	1.657	0.710	0.560	0.150
⋮						
25	7.321	2.706	1.894	0.812	0.732	0.080
⋮						
100	8.962	2.994	2.096	0.898	0.896	0.002
⋮						
∞	9.000	3.000	2.100	0.900	0.900	0.000

Alternatively we can use the equation:
$sf(k^*) = \delta k^*$
to calculate the *steady state* capital stock and the corresponding output, consumption, and investment.

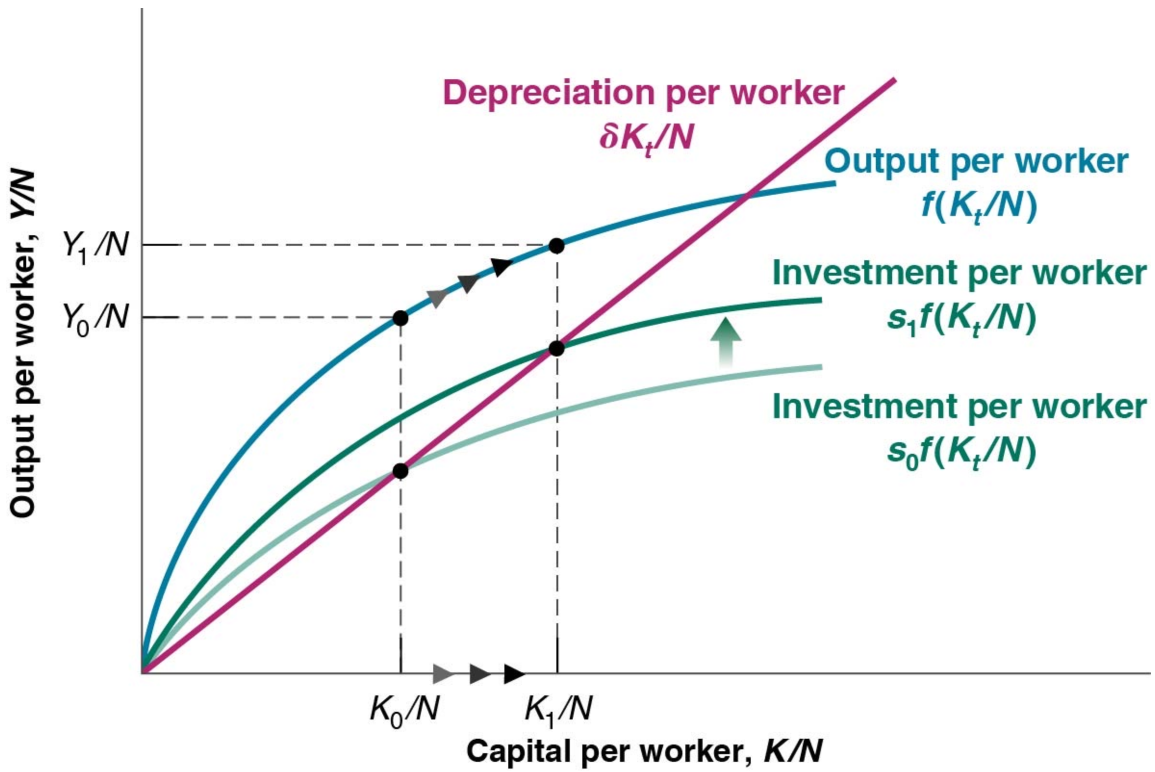
11.2 The Implications of Alternative Saving Rates (4 of 10)

- The saving rate has no effect on the long-run growth rate of output per worker, which is equal to zero.
- The saving rate determines the level of output per worker in the long run.
- An increase in the saving rate will lead to higher growth of output per worker for some time, but not forever.

11.2 The Implications of Alternative Saving Rates (5 of 10)

Figure 11.3 The Effects of Different Saving Rates

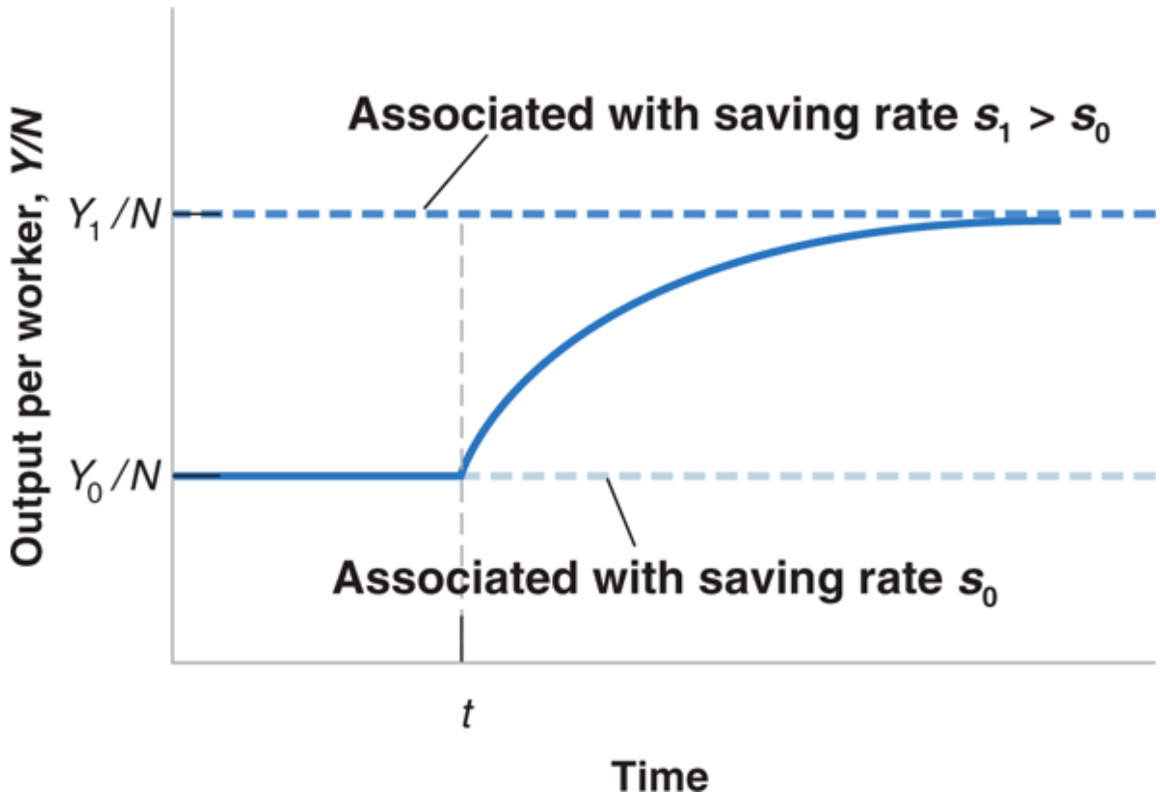
A country with a higher saving rate achieves a higher steady-state level of output per worker.



11.2 The Implications of Alternative Saving Rates (6 of 10)

Figure 11.4 The Effects of an Increase in the Saving Rate on Output per Worker in an Economy Without Technological Progress

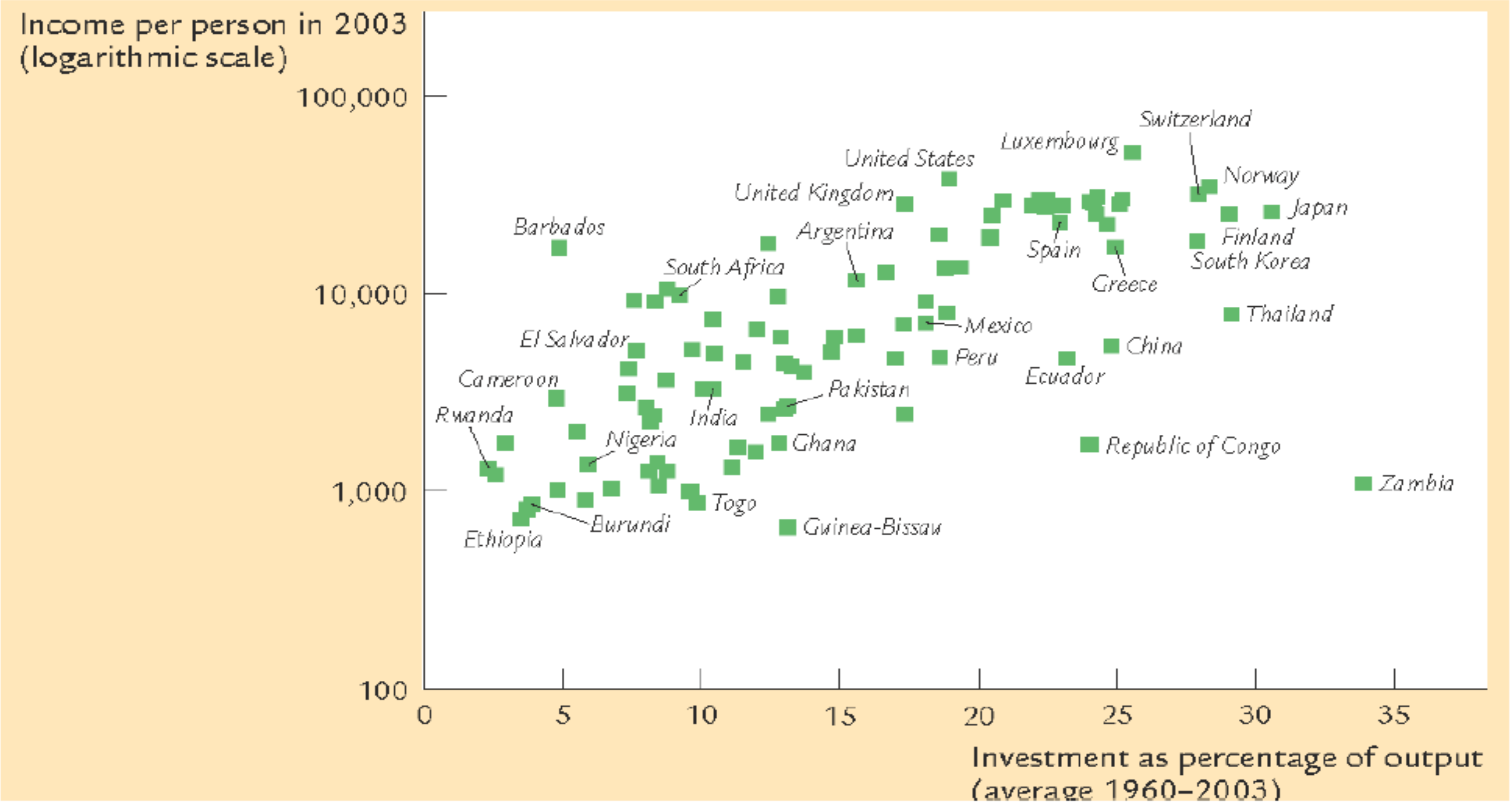
An increase in the saving rate leads to a period of higher growth until output reaches its new higher steady-state level. (only level effect, no growth effect)



Pearson

Saving and Investment

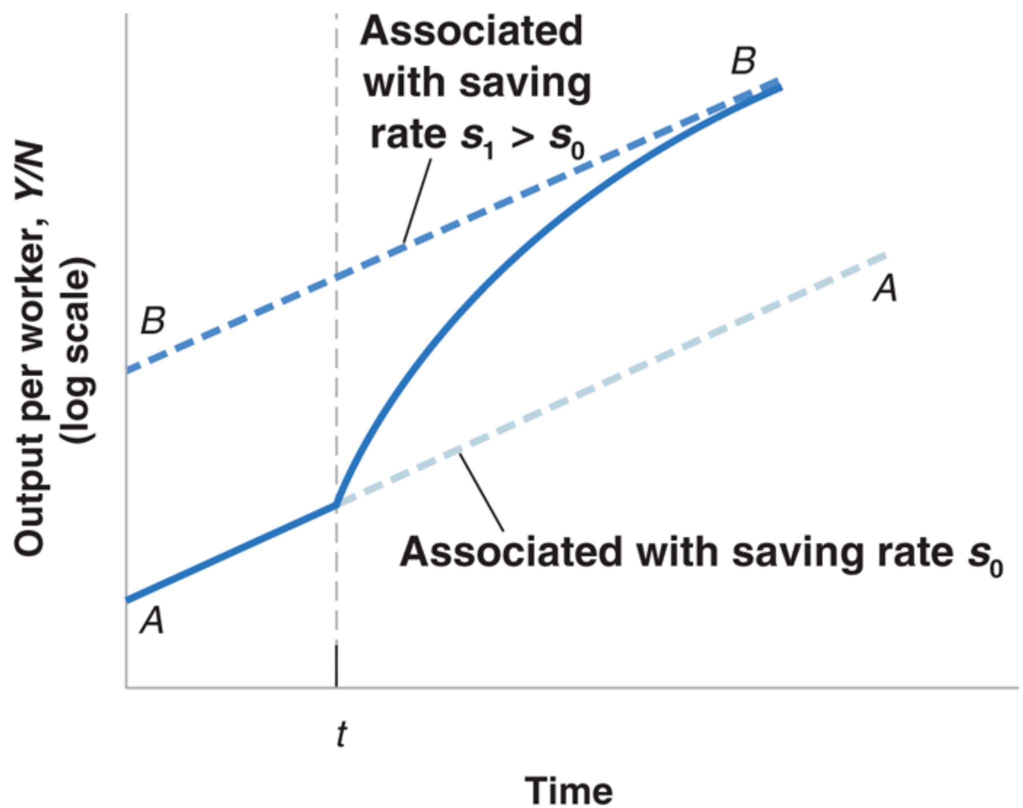
The Solow Model predicts that countries with high saving rate have higher steady state capital stock per capita and hence higher income per capita. Is this really the case?



11.2 The Implications of Alternative Saving Rates (7 of 10)

Figure 11.5 The Effects of an Increase in the Saving Rate on Output per Worker in an Economy with Technological Progress

An increase in the saving rate leads to a period of higher growth until output reaches its new higher steady-state level.



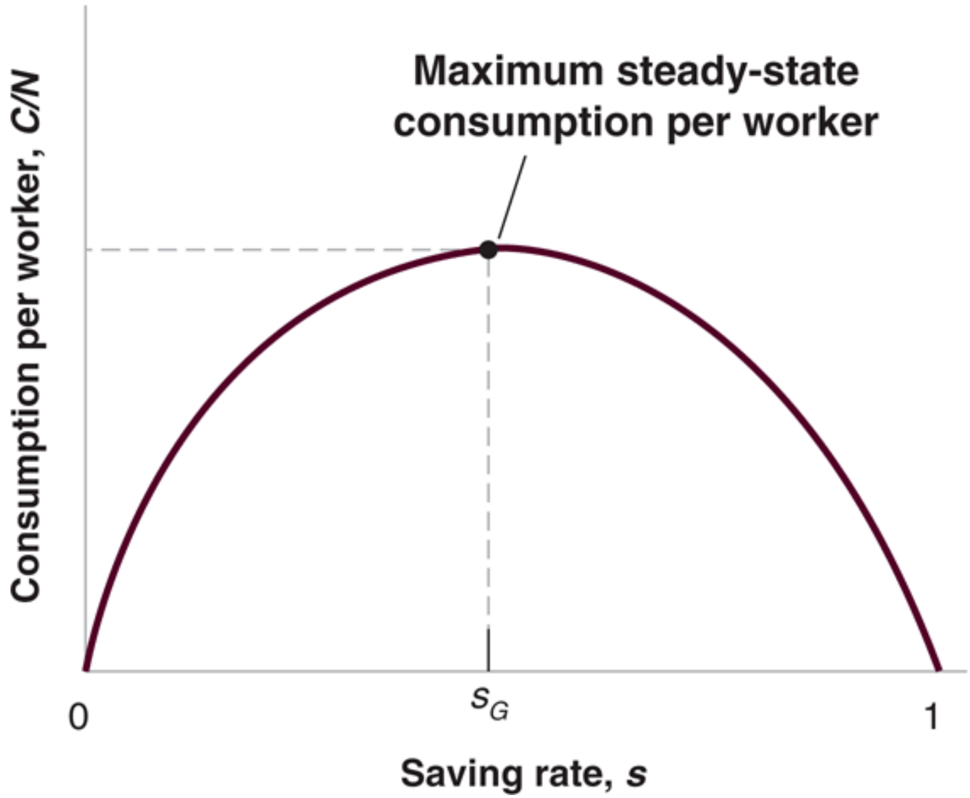
11.2 The Implications of Alternative Saving Rates (8 of 10)

- What matters to people is not how many is produced, but how much they consume.
- Governments can affect the saving rate by:
  – changing public saving (budget surplus)
  – using taxes to affect private saving
- **Golden-rule level of capital:** The level of capital associated with the saving rate that yields the highest level of consumption in steady state.

11.2 The Implications of Alternative Saving Rates (9 of 10)

Figure 11.6 The Effects of the Saving Rate on Steady-State Consumption per Worker

An increase in the saving rate leads to an increase, then to a decrease in steady-state consumption per worker.

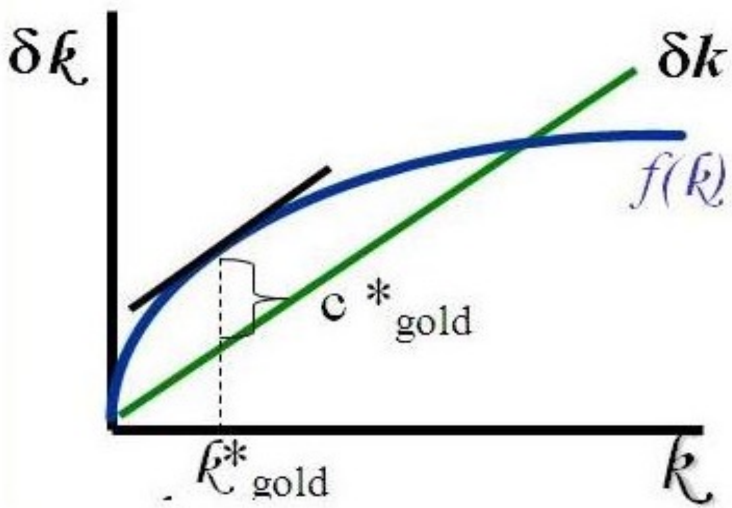


11.2 The Implications of Alternative Saving Rates (10 of 10)

- For a saving rate between zero and the golden-rule level, a higher saving rate leads to higher capital per worker, higher output per worker and higher consumption per worker.
- For a saving rate greater than the golden-rule level, a higher saving rate still leads to higher capital per worker and output per worker, but lower consumption per worker.
- An increase in the saving rate leads to lower consumption for some time but higher consumption later.

6. The Golden Rule Level of Capital

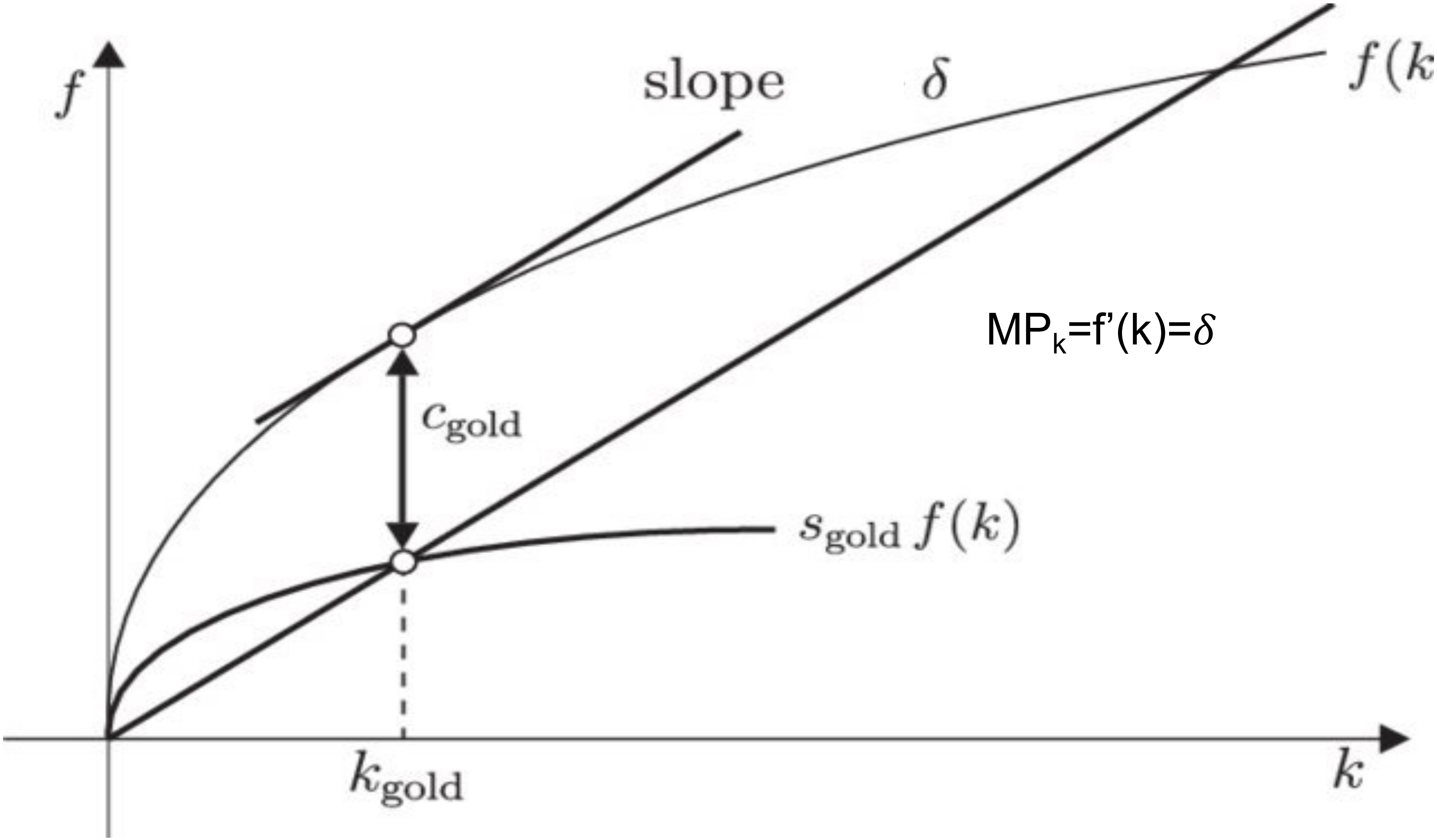
- *The steady-state value of k that maximizes consumption* is called the **Golden Rule Level of Capital**. (k^*_{gold})
- national income accounts identity: y = c+i → c = y-i
- Substitute steady-state values: Steady-state output per worker is **f (k*)**; capital stock is not changing in the steady state, investment =depreciation δk^*. → steady-state consumption per worker $\boldsymbol{c^* = f(k^*) - \delta k^*}$



In the k^*_{gold} the slope of the production function (MPK) is equal to the slope of the depreciation function (δ) → **At the Golden Rule level of capital**, the marginal product of capital equals the depreciation rate.

$$\mathbf{MPK=\delta}$$

Golden Rule Saving Rate



$MP_k = f'(k) = \delta$

Finding the Golden Rule Steady State: A Numerical Example

Assumptions: $y = \sqrt{k}$; $\delta = 0.1$

s	k^*	y^*	δk^*	c^*	MPK	$MPK - \delta$
0.0	0.0	0.0	0.0	0.0	∞	∞
0.1	1.0	1.0	0.1	0.9	0.500	0.400
0.2	4.0	2.0	0.4	1.6	0.250	0.150
0.3	9.0	3.0	0.9	2.1	0.167	0.067
0.4	16.0	4.0	1.6	2.4	0.125	0.025
0.5	**25.0**	**5.0**	**2.5**	**2.5**	**0.100**	**0.000**
0.6	36.0	6.0	3.6	2.4	0.083	–0.017
0.7	49.0	7.0	4.9	2.1	0.071	–0.029
0.8	64.0	8.0	6.4	1.6	0.062	–0.038
0.9	81.0	9.0	8.1	0.9	0.056	–0.044
1.0	100.0	10.0	10.0	0.0	0.050	–0.050

FOCUS: Social Security, Saving, and Capital Accumulation in the United States

- Social Security, introduced in 1935, has led to a lower U.S. saving rate and thus lower capital accumulation and lower output per person in the long run.
- Social Security is a **pay-as-you-can system** that taxes workers and redistributes the tax contributions as benefits to current retirees, resulting in lower private saving as workers anticipate receiving benefits when they retire.
- An alternative is a **fully-funded** system that pays back the principal plus interest to the workers when they retire, resulting in lower private saving but higher public saving as the System invests their contributions in financial assets.

11.3 Getting a Sense of Magnitudes (1 of 6)

- Assume the production function *f*:

$$Y = \sqrt{\mathrm{K}}\,\sqrt{\mathrm{N}} \qquad (11.6)$$

- so that equation (11.3) becomes:

$$\frac{K_{t+1}}{N} - \frac{K_t}{N} = s\sqrt{\frac{Kt}{N}} - \delta\,\frac{K_t}{N} \qquad (11.7)$$

- which describes the evolution of capital over time.

11.3 Getting a Sense of Magnitudes (2 of 6)

- Equation (11.7) implies that capital per worker in the steady state (K^*/N) becomes:

$$\frac{K^*}{N} = \left(\frac{s}{\delta}\right)^2 \tag{11.8}$$

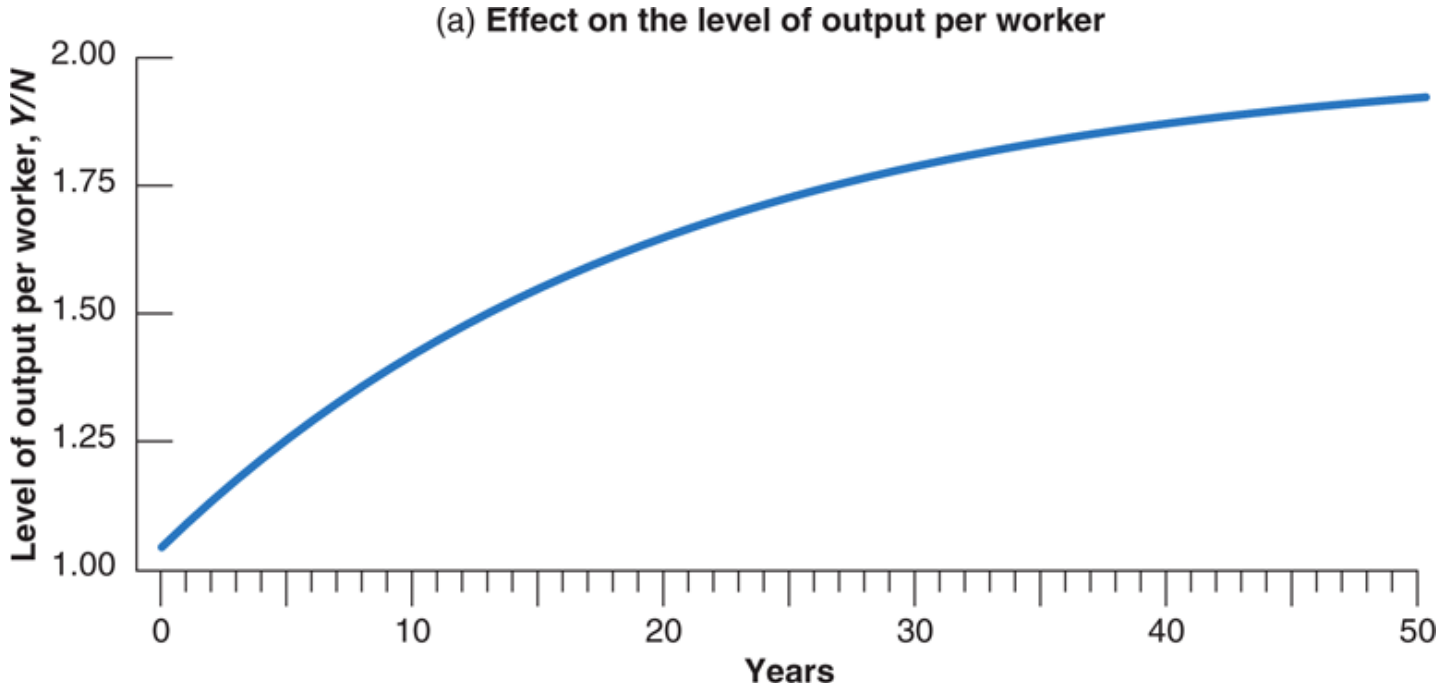
- Combining equations (11.6) and (11.8) gives the steady state output per worker:

$$\frac{Y^*}{N} = \sqrt{\frac{K^*}{N}} = \sqrt{\left(\frac{s}{\delta}\right)^2} = \frac{s}{\delta} \tag{11.9}$$

- In the long run, output per worker doubles when the saving rate doubles.

11.3 Getting a Sense of Magnitudes (3 of 6)

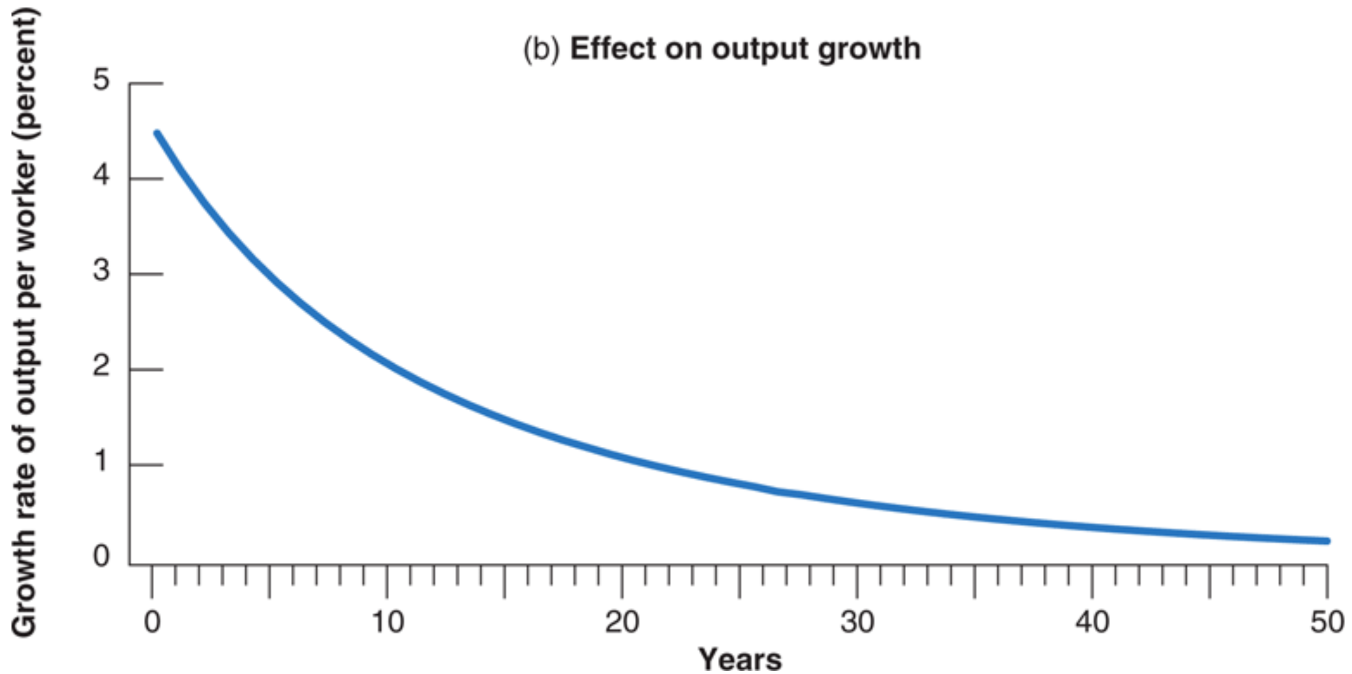
Figure 11.7(a) The Dynamic Effects of an Increase in the Saving Rate from 10% to 20% on the Level and the Growth Rate of Output per Worker



It takes a long time for output to adjust to its new higher level after an increase in the saving rate. Put another way, an increase in the saving rate leads to a long period of higher growth.

11.3 Getting a Sense of Magnitudes (4 of 6)

Figure 11.7(b) The Dynamic Effects of an Increase in the Saving Rate from 10% to 20% on the Level and the Growth Rate of Output per Worker



It takes a long time for output to adjust to its new higher level after an increase in the saving rate. Put another way, an increase in the saving rate leads to a long period of higher growth.

- In the steady state, consumption per worker is:

$\hat{i} = S = \beta y = \beta f(k^*)$ s.t. $\beta f(k^*) = \delta k^*$

$\Delta K = \beta f(k^*) - \delta k^*$

$$\frac{C}{N} = \frac{Y}{N} - \delta \frac{K}{N}$$

$c = y - \hat{i} = \beta f(k^*) - \delta k^*$

- Given equations (11.8) and (11.9), the steady-state consumption per worker is:

$$\frac{C}{N} = \frac{s}{\delta} - \delta \left(\frac{s}{\delta}\right)^2 = \frac{s(1-s)}{\delta}$$

- Table 11.1 gives the steady-state values of capital per worker, output per worker and consumption per worker for different saving rates (given δ=10%)

Pearson

11.3 Getting a Sense of Magnitudes (6 of 6)

Table 11.1 The Saving Rate and the Steady-State Levels of Capital, Output, and Consumption per Worker

Saving Rate *s*	Capital per Worker *K* / *N*	Output per Worker *Y* / *N*	Consumption per Worker *C* / *N*
0.0	0.0	0.0	0.0
0.1	1.0	1.0	0.9
0.2	4.0	2.0	1.6
0.3	9.0	3.0	2.1
0.4	16.0	4.0	2.4
0.5	25.0	5.0	2.5
0.6	36.0	6.0	2.4
. . .	. . .	. . .	. . .
1.0	100.0	10.0	0.0

Population growth in Solow' model

- Assume that the population--and labor force-- grow at rate $\boldsymbol{n}$. ($\boldsymbol{n}$ is exogenous)

$$\frac{\Delta L}{L} = n$$

- $(\delta + n)k$ = **break-even investment**, the amount of investment necessary to keep k constant.

Break-even investment includes:

Δk =sf(k)- $(\delta + n)k$

$(\delta + n)$ is the effective depreciation rate

- δk to replace capital as it wears out
- $n k$ to equip new workers with capital

  (otherwise, k would fall as the existing capital stock would be spread more thinly over a larger population of workers)

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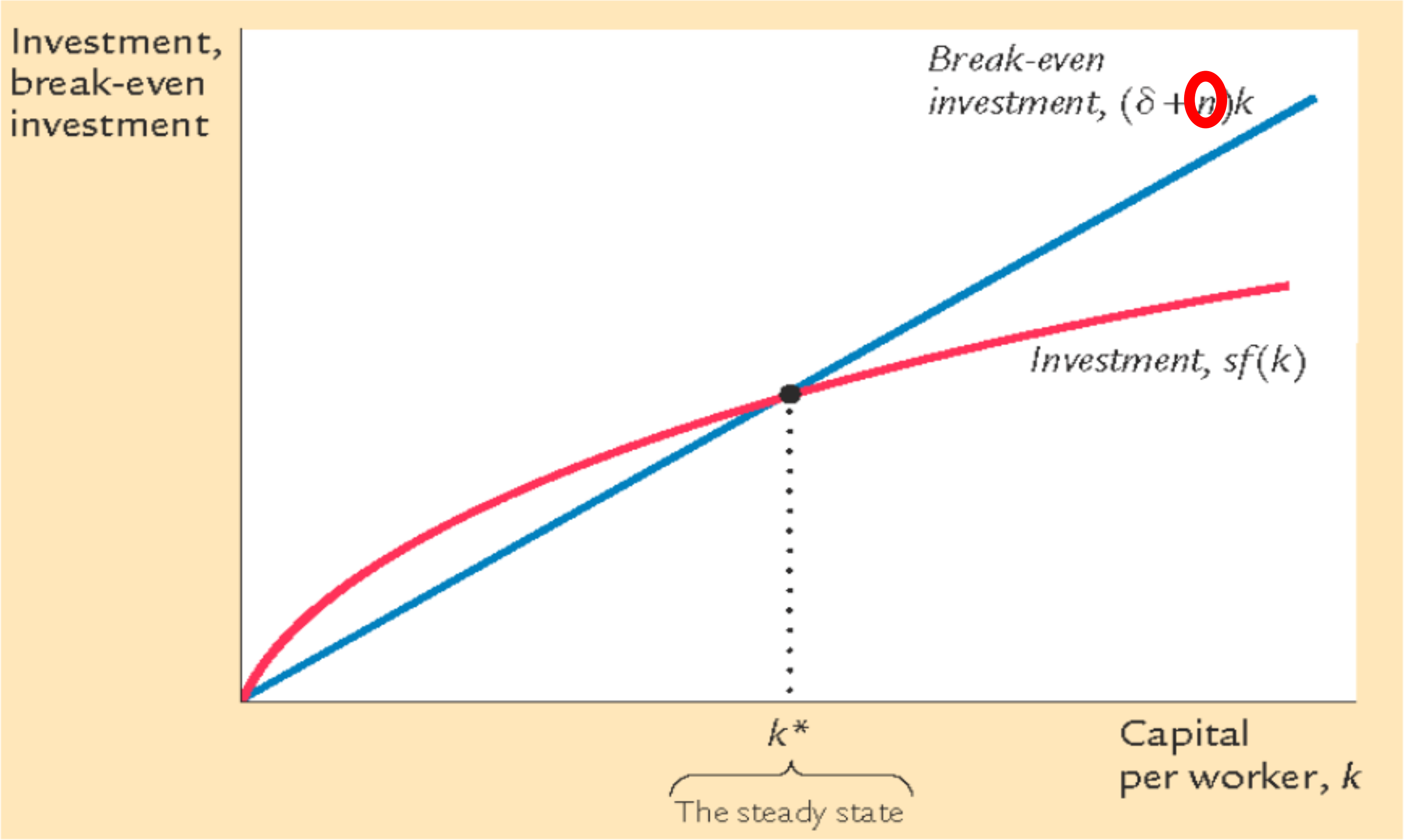
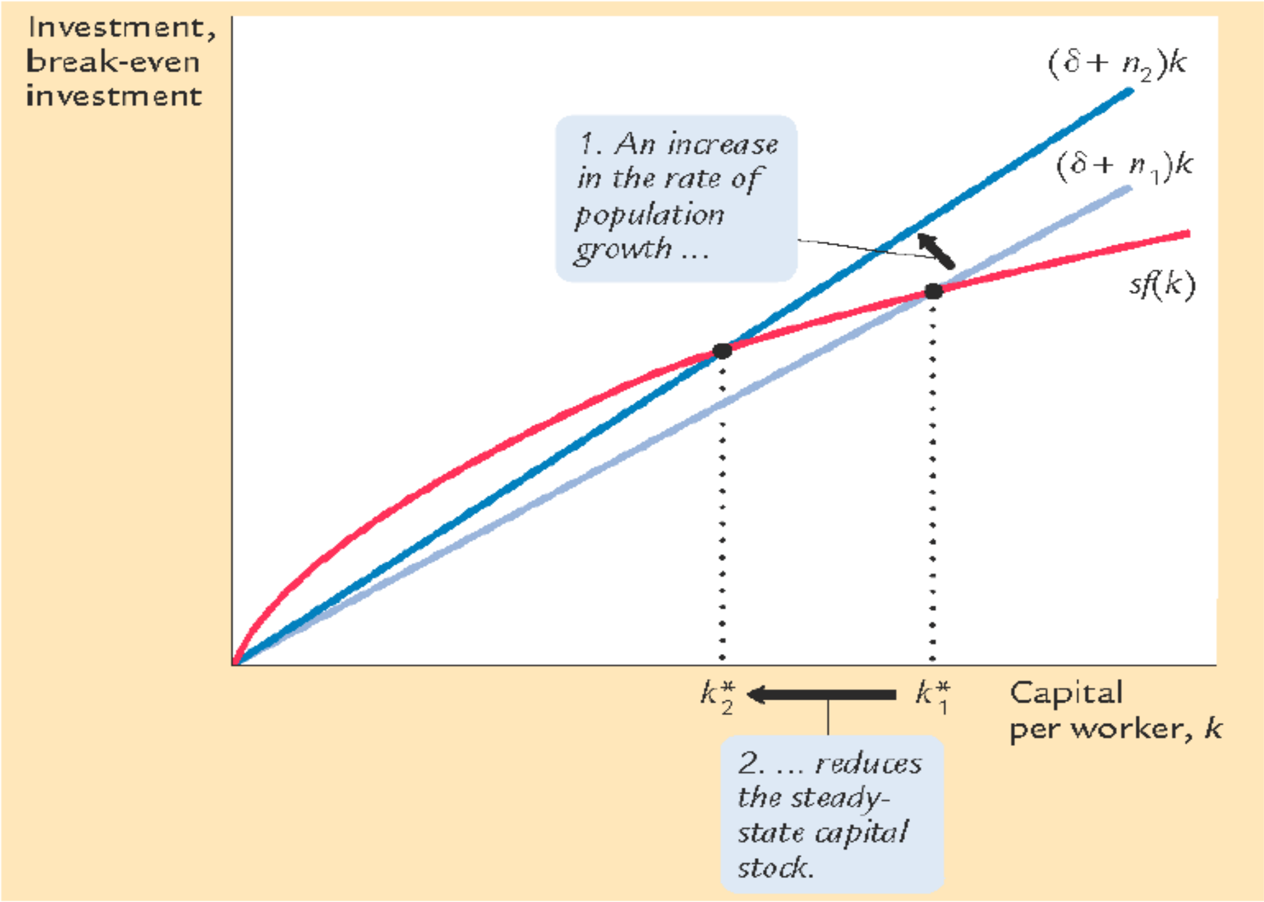
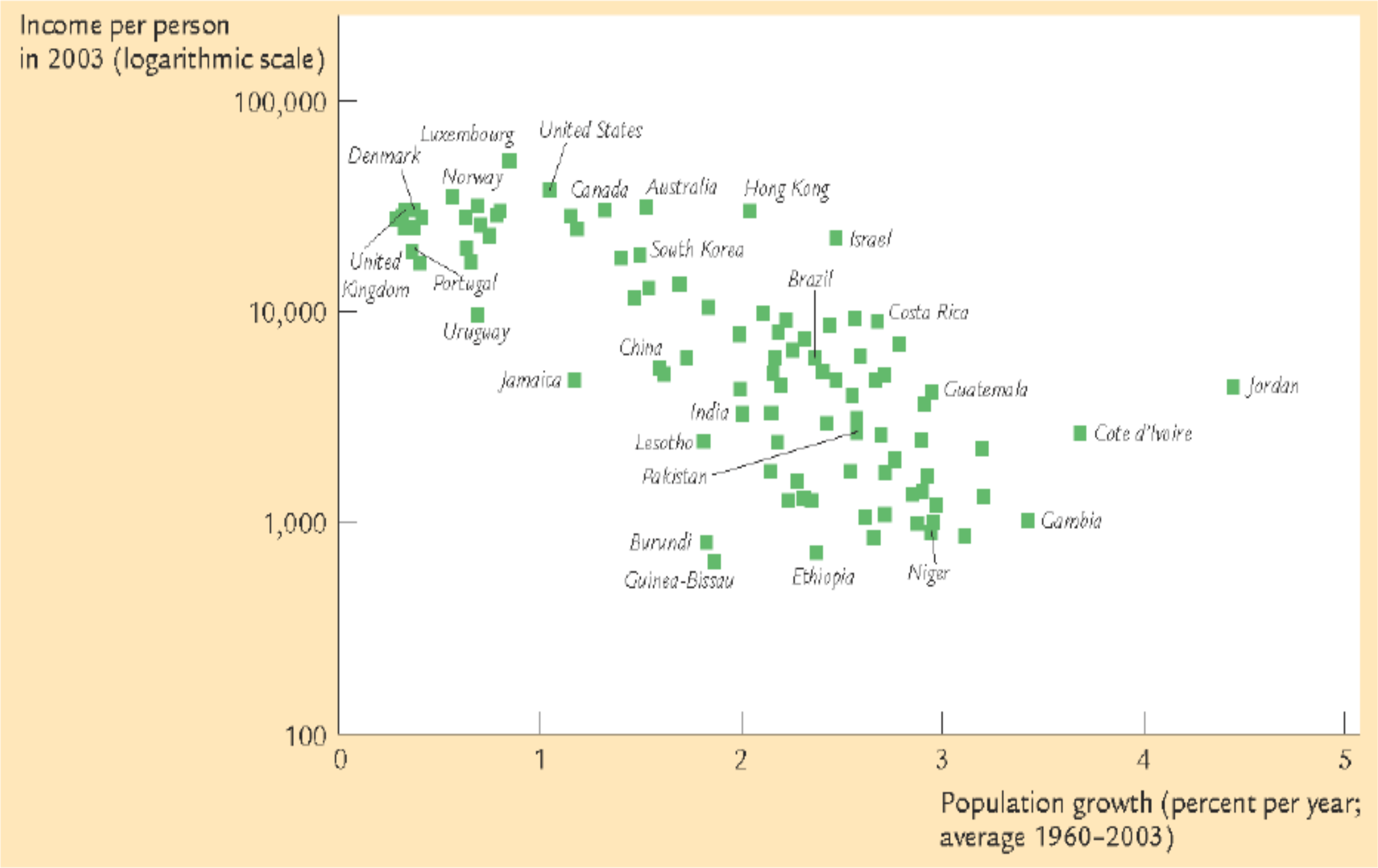
Population growth in Solow' model



Effect of population growth

The Solow model predicts that countries with higher population growth rates will have lower levels of capital and income per worker in the long run. Is this the case?




Income per person in 2003 (logarithmic scale)
100,000
10,000
1,000
100
0
1
2
3
4
5
Population growth (percent per year; average 1960–2003)
Luxembourg
United States
Denmark
Norway
Canada
Australia
Hong Kong
Israel
South Korea
United Kingdom
Portugal
Brazil
Costa Rica
Uruguay
China
Jamaica
Guatemala
Jordan
India
Lesotho
Cote d'Ivoire
Pakistan
Gambia
Burundi
Guinea-Bissau
Ethiopia
Niger

FOCUS: Nudging US Households to Save More

- US households save little: The household saving rate, defined as the ratio of household saving to household disposable income, has averaged 6% since 2000.
- Successive governments have tried to increase the saving rate by giving tax breaks, which made saving more attractive, but with limited success.
- Nobel Laurette Richard Thaler, considered one of the founders of behavioral economics, designed a program to increase saving based on human behavior.
- Thaler and Benartzi developed the ‘Save More Tomorrow’ approach which has helped approximately 15 million Americans significantly boost their savings rate.

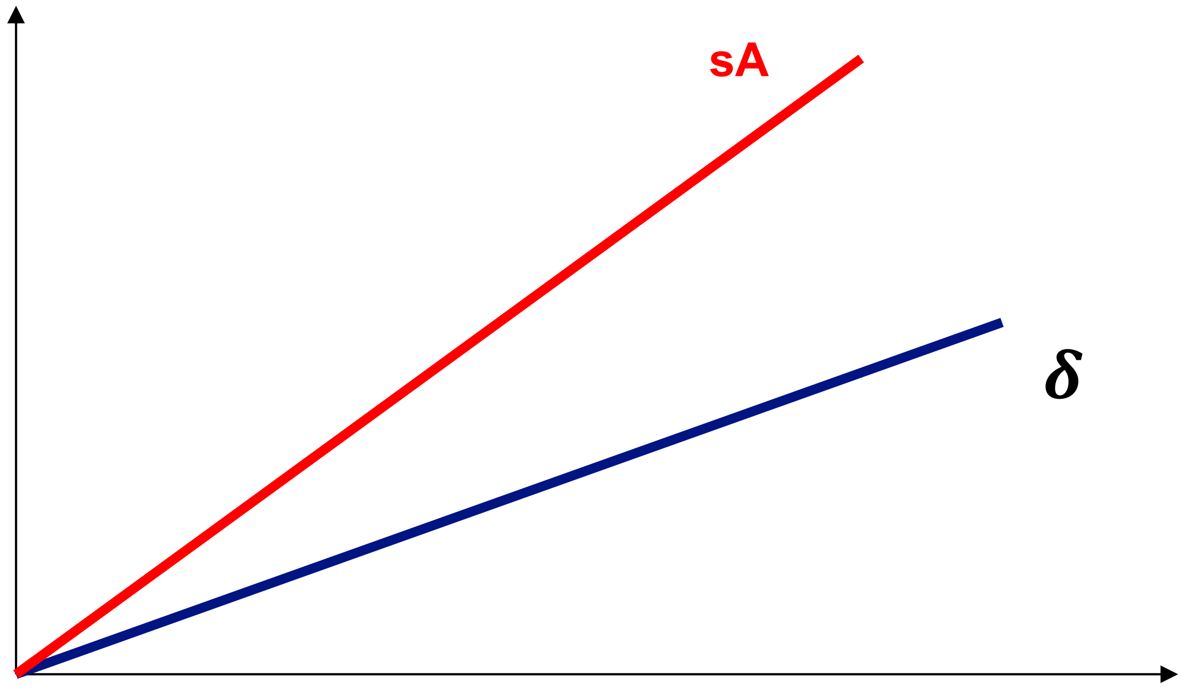
Exogenous Technology Change

- *Labor-augmented technology progress*
- $Y = K^{\alpha}(AN)^{1-\alpha}$ and $\frac{\Delta A}{A} = g_A,\ \frac{\Delta N}{N} = n$
- Let y=Y/AN, k=K/AN, measured by per capita in efficiency unit
- $y = k^{\alpha}$
- Δk =sf(k)- $(\delta + n + g_A)k$
- In the long run, $\frac{\Delta k}{k} = 0, \textit{which implies}\ \frac{K}{AN}\ \textit{is constant}$
- Growth rate of K, $\frac{\Delta K}{K} = g_A$+n, so is the aggregate output growth
- Growth rate of output (or capital, consumption) per capita= g_A
- There is a balanced growth path for the economy ($g_{Y/N}$= $g_{K/N}$= $g_{C/N}$= g_A)

Pearson

Ak model, i.e., $\alpha = 1$, C.R.S on k or MPK is constant

- Δk =sf(k)- δk=sAk - δk
- $\frac{\Delta y}{y} = \Delta k/k$ =sf(k)/k- δ=sA - δ >0
-
-



11.4 Physical versus Human Capital (1 of 2)

- **Human capital** (*H*): The set of skills of the workers in the economy built through education and on-the-job training.
- The production function with human capital:

$$\frac{Y}{N} = f\left(\underset{(+,}{\frac{K}{N}}, \underset{+)}{\frac{H}{N}}\right) \qquad (11.10)$$

- As for physical capital (*K*) accumulation, countries that save more or spend more on education can achieve higher steady-state levels of output per worker.

- Both K and H can accumulate over time and with C.R.S on K and H

Various types of capital: Human capital, social capital,…../

- $y = Af(\sum_1^n k_i) = A\sum k_i^{\alpha i}$, where *y=Y/L, k=K/L, output per capita*
- $\sum_1^n \alpha_i = 1$
- $y = A \cdot K$

Infrastructure Building as an engine of growth

Private firm's production function

- $Y_i = AF(K_i, L_i) = AK_i^{\alpha} L_i^{1-\alpha}$, α=capital income share≈1/3, A is technology factor or efficiency level
- $y_i = Af(k_i) = Ak_i^{\alpha}$, where *y=Y/L, k=K/L, output per capita*
- $(y = \sum_1^n y_i = A \sum_1^n k_i^{\alpha} = A \cdot K^{\alpha})$
- *Capital accumulation drives output growth but subject to diminishing return resulting zero growth in the long run*
- *Infrastructure* $A(K) = K = \sum_1^n k_i$

Aggregate output of the economy

- $y = \sum_1^n y_i = A \sum_1^n k_i^{\alpha} = K \cdot K^{\alpha} = K^{1+\alpha}$, *Increasing return on capital accumulation leads to unlimited growth*

11.4 Physical versus Human Capital (2 of 2)

- **Models of endogenous growth:** Steady-state growth in output per worker depends on variables such as the saving rate and the rate of spending on education, even without technological progress.
- However, the current consensus is that given the rate of technological progress, higher rates of saving or spending on education do not lead to a permanently higher growth rate.

APPENDIX: The Cobb-Douglas Production Function and the Steady State (1 of 2)

- The Cobb-Douglas production function:

$$Y = K^{\alpha} N^{1-\alpha} \qquad (11.A1)$$

which gives a good description of the relation between output, physical capital, and labor in the United States from 1899 to 1922.

- In steady state, saving per worker must be equal to depreciation per worker, implying that:

$$s(K^*/N)^{\alpha} = \delta(K^*/N)$$

where K^* is the steady-state level of capital.

APPENDIX: The Cobb-Douglas Production Function and the Steady State (2 of 2)

- The preceding expression can be rewritten as:

$$s = \delta(K^*/N)^{1-\alpha}$$

- The steady-state level of capital per worker becomes:

$$(K^*/N) = (s/\delta)^{\alpha/(1-\alpha)}$$

- If $\alpha = 0.5$, then:

$$K^*/N = s/\delta$$

which implies that a doubling of the saving rate leads to a doubling in steady-state output per worker.